

What are the risks in IoT?

[Source: <https://securityscorecard.com/blog/internet-of-things-threats-and-risks/>]

The Internet of Things (IoT) is a network of intertwined devices, software, sensors, and other ‘things’ which enable the world to be connected throughout physical space. This can include business software, smart home devices, care monitoring systems, mobile phones, or driverless trucks, and can be as small as a thumb drive to the size of a train. All of these things communicate with each other without the need for human interaction. This spider web of connectivity is fascinating but poses serious danger to information security.

The internet of things (IoT) is a highly developed space that is home to a vast amount of sensitive data, making it a very attractive target for cybercriminals. Threats and risks continue to evolve as hackers come up with new ways to breach unsecured systems — posing a threat to the ecosystem itself.

Let’s take a look at the leading threats and risks to the IoT and the associated vulnerabilities that must be secured.

Why does IoT security matter?

The ubiquity of smart devices is a major consideration for cybersecurity. Vulnerability in one of these IoT devices can lead to costly data breaches and affect an entire organization’s productivity. IoT security is also important as it keeps data secure.

Smart devices can house large amounts of sensitive data, all of which falls under specific cybersecurity regulations. If this information is not secured, there could be legal ramifications if the data is compromised.

Exploring the IoT attack surface

A business’s attack surface is the sum of vulnerabilities that are currently present on its network, both physical and digital. This can be vulnerabilities from within endpoint devices (computers, tablets, etc.) or from the software and hardware used to conduct business.

While each device is typically protected through security software, they are still apt to a series of added threats and vulnerabilities through their connection to the IoT.

The Open Web Application Security Project (OWASP) provides a broad consensus of the current threats and vulnerabilities within the surfaces, which are outlined as OWASP Top Ten.

Devices

Devices inevitably have vulnerabilities embedded within their memory systems, physical and web interface, network services, and firmware. This allows hackers to easily exploit systems within the devices’ outdated components and insecure default settings with update mechanisms. When managing vulnerabilities throughout your network’s devices, continuous monitoring is essential.

Communication channels

Attacks can originate from the channels that connect IoT devices. This presents serious threats to the security of the entire system and creates a potential for spoofing and Denial-of-Service (DoS) attacks. These threats and attacks lay the foundation for an unstable network surface.

Applications and software

Each application and software presents risk and many web applications and APIs do not protect sensitive data adequately. This data can be anything from financial intelligence to healthcare information. A breach of these types of information can result in identity theft, credit card fraud, and exposure of confidential information all because a web application isn't properly secured or patched on a consistent basis.